Radboud Universiteit

Bayesian regression for language scientists

Lecture 5

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Outline

- Bayes factors
- Best practices
- Reporting the results
- Your ideas for applying Bayesian regression
- Q & A
- Survey

Bayes Factors

- We often want to know which of two hypotheses is best supported by the data.
- We can use **Bayes factor (BF)** to answer this question.

$$BF_{01} = \frac{P\,(H_0|data)}{P\,(H_1|data)} = \frac{P\,(data|H_0)\,P\,(H_0)/P\,(data)}{P\,(data|H_1)\,P\,(H_1)/P(data)}$$

$$= \frac{P\,(data|H_0)}{P\,(data|H_1)}$$

Ratio of Marginal likelihoods

Interpretation

- $BF_{01} < 1$ represents **relative** evidence in favour of the alternative hypothesis H_1 or the model that captures it
- $BF_{01} > 1$ shows **relative** evidence in favour of the null hypothesis H_0 or the model that captures it.

Evidence	BF in favour of the null hypothesis	BF in favour of the alternative hypothesis
Anecdotal	Between 1 and 3	Between 1/3 and 1
Moderate	Between 3 and 10	Between 1/10 and 1/3
Strong	> 10	< 1/10

Pros and cons of BF

- Pros:
  - BF can measure the strength of support for the null hypothesis (compared to the alternative hypothesis).
  - This can lead to fewer studies with null results ending up in a drawer.
- Cons:
  - Very sensitive to the choice of priors.
  - To obtain a meaningful BF, you need strongly informative, well-motivated priors, which are not always easy to define.

Example: Cookies

- Let us compare the evidence for the model with only the intercept (model 0) and the model with cookies as a predictor (model 1).
- First, let us use flat (default) priors for the effect of cookies.
- We need to save all internal parameters, which are by default discarded to make the model object smaller.

```
m_brm_flat0 <- brm(gain ~ 1, data = df_cookies, cores =
4, save_pars = save_pars(all = TRUE))

m_brm_flat1 <- brm(gain ~ cookies, data = df_cookies,
cores = 4, save_pars = save_pars(all = TRUE))
```

Log-marginal likelihood

- Model 0:

```
logml_flat0 <- bridge_sampler(m_brm_flat0)
logml_flat0
Bridge sampling estimate of the log marginal likelihood: -18.56785
Estimate obtained in 5 iteration(s) via method "normal".
```

- Model 1:

```
logml_flat1 <- bridge_sampler(m_brm_flat1)
logml_flat1
Bridge sampling estimate of the log marginal likelihood: -17.72617
Estimate obtained in 5 iteration(s) via method "normal".
```

BF = Ratio of marginal likelihoods

```
exp(logml_flat0$logml)/exp(logml_flat1$logml)
[1] 0.4309844

bayes_factor(logml_flat0, logml_flat1)
Estimated Bayes factor in favor of x1 over x2: 0.43098
```

Anecdotal evidence in favour of Model 1!

Changing the order: Model 1 vs. Model 0

```
bayes_factor(logml_flat1, logml_flat0)
Estimated Bayes factor in favor of x1 over x2: 2.32027
```

Same interpretation!

Weakly informative priors

```
my_priors_weak <- prior(normal(0, 10), class = "b")
m_brm_weak0 <- brm(gain ~ 1, data = df_cookies, cores =
4, save_pars = save_pars(all = TRUE))
m_brm_weak1 <- brm(gain ~ cookies, data = df_cookies,
cores = 4, prior = my_priors_weak, save_pars =
save_pars(all = TRUE))

bayes_factor(m_brm_weak0, m_brm_weak1)
```

```
Estimated Bayes factor in favor of m_brm_weak0 over
m_brm_weak1: 10.86189
```

Strong evidence in favour of Model 0!

Strong specific priors

```
my_priors_strong <- prior(normal(0.10, 0.1), class =
"b")
m_brm_strong0 <- brm(gain ~ 1, data = df_cookies, cores
= 4, save_pars = save_pars(all = TRUE))
m_brm_strong1 <- brm(gain ~ cookies, data = df_cookies,
cores = 4, prior = my_priors_strong, save_pars =
save_pars(all = TRUE))

bayes_factor(m_brm_strong1, m_brm_strong0)
```

```
Estimated Bayes factor in favor of m_brm_strong1 over
m_brm_strong0: 8.11258
```

Moderate evidence in favour of Model 1!

Exercise

- Work in groups and fit mixed-effects models **with and without University** on the simulated data from Day 2:

```
brm(Nwords ~ (1|Student_ID) + (1|Topic_ID) + Vocab_Score + University,
data = df, cores = 4, save_pars = save_pars(all = TRUE))
```

  - Group 1 uses default flat priors for both fixed effects
  - Group 2 uses normal(0, 10) priors for both fixed effects
  - Group 3 uses strong informative priors:

```
my_prior_university <- prior(normal(8, 1), class = "b", coef =
    "University1")
my_prior_vocab <- prior(normal(0.1, 0.1), class = "b", coef =
"Vocab_Score")
```

  Tip: use `brm(…, prior = my_prior_university + my_prior_vocab)`

- Compute Bayes Factor and report it.

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Best practices

- Think carefully about priors and describe them in detail.
- Perform a prior sensitivity analysis and report the results.
- Perform Bayesian chain convergence diagnostics (R-hats, effective sample sizes, visual diagnostics). The chains should mix well.
- Provide as much information about the posterior distribution of a coefficient as possible - not only the mean or median, but also credible intervals - ideally with different quantiles. Inspect the shape of the posteriors, trying to avoid multimodal distributions.
- Perform posterior predictive checks, which are often done with the help of simulation. One simulates new data conditional on the posterior distribution and checks how much this data resembles the original data.

Replicability

- The usual method with `set.seed()` cannot control MCMC in Stan.
- However, you can specify the random seed in `brm(… , seed = n)`, choosing any number.
- The estimates will be stable.

Multiple tests

- In frequentist statistics, performing multiple tests on the same data is dangerous because it increases the chances of Type I error.
  - statistical corrections for multiple comparisons between groups
  - Gelman et al. (2009): if we perform a Bayesian hierarchical analysis, we do not need to worry about multiple comparisons (based on a simulation study).

- Frank Harrell (blog posts and comments):
  - In the frequentist framework, performing multiple tests increases the number of opportunities for the observed data to produce extreme results purely by chance. We focus on the probability of extreme data, which we should reduce. There are no built-in constraints on extreme data, so we need some additional adjustments.
  - In Bayesian statistics, this is not the case. We focus on the probability of parameters given data. Data is a given. Constraints on the probability of extreme values are introduced through the prior distribution, so no further adjustments are needed.

Diagnostics checklist

- Chain diagnostics: Do the chains converge? What's the R-hat values? Are ESS enough?
- Is data generated by the model similar to the actual data?
- Additionally:
  - Does the model underfit or underfit the data?
  - Are there weird observations that affect the posteriors too strongly?
  - Is there multicollinearity?
  - Are there non-linear effects that we have failed to capture?

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Marginal and conditional R^2

- Function `bayes_R2()` in brms provides the so-called conditional R^2 – based on both random and fixed effects.

```
bayes_R2(mm_brm)

    Estimate   Est.Error      Q2.5     Q97.5
R2 0.8771563 0.003220434 0.8703944 0.8829212
```

- For a Bayesian mixed-effects model, it is also possible to compute the marginal R^2: how much variation is explained by the fixed effects only.

```
library(performance)
r2_bayes(mm_brm)
# Bayesian R2 with Compatibility Interval
  Conditional R2: 0.877 (95% CI [0.871, 0.883])
     Marginal R2: 0.132 (95% CI [0.050, 0.222])
```

Overfitting

- An overfitting model provides a good fit to the data, but the model is useless if we take a new sample.
- An extreme example: a model with the same number of parameters as the number of observations.
- To detect overfitting, one can use cross-validation.
    - A luxury: using the model to predict the response on a new sample.
    - A more realistic option:
        - draw subsamples from the data
        - compute predicted values of the response based on the original model
        - compare goodness-of-fit statistics (error, variance, etc.) for the whole sample and subsample
        - If the statistic is much better for the whole sample than for the subsample, there is risk of overfitting
        - Repeat several times with different random seeds.

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Influential observations

- Method for detecting influential observations: Leave-One-Out cross-validation.
  - For each observation, compute a Pareto shape parameter *k*, which represents by how much the posteriors change if one removes this data point in comparison with the full data. Influential observations will have high *k* values. As a rule of thumb, large Pareto *k* values (higher than 0.7) mean that the model is not robust.

```
check_outliers(brm_mm) #from package "performance"
```

```
OK: No outliers detected.
- Based on the following method and threshold: pareto (0.7).
- For variable: (Whole model)
```

Some technical details

- LOO criterion is important for estimating the model's predictive accuracy
  - Ideally, we should leave out every observation and then refit the model.
  - But this is VERY impractical, so the statistics are computed using an efficient approximation.
- Some observations may produce high **pareto_k values** (> 0.7). Leaving them out affects the posteriors strongly. This can produce inaccuracte LOO estimates.
  - Causes: influential observations, model specification, random variation…
- To correct this, we can use the moment matching method as a patch. It is more computationally intensive, but still more efficient than refitting the model for every problematic observation.
- In order to be able to use this method, all parameters for inner computations which are normally discarded, should be saved by adding save_pars = save_pars(all = TRUE). Note that this increases the size of the model object.

Diagnostics checklist

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Multicollinearity

- VIF (variance inflation factor) is a measure for diagnostics of multicollinearity of model terms. They are used both in the frequentist and Bayesian approaches.
- A VIF < 5 means indicates a low correlation of that predictor with other predictors.
- A VIF > 5 and especially 10 indicates multicollinearity.

```
check_collinearity(mm_brm) #from package "performance"
```

```
# Check for Multicollinearity

Low Correlation

       Term  VIF        VIF 95% CI Increased SE Tolerance
 Vocab_Test 1.00 [1.00, 1.21e+09]         1.00      1.00
 University 1.00 [1.00, 1.21e+09]         1.00      1.00
 Tolerance 95% CI
     [0.00, 1.00]
     [0.00, 1.00]
```

Diagnostics checklist

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Non-linear effects

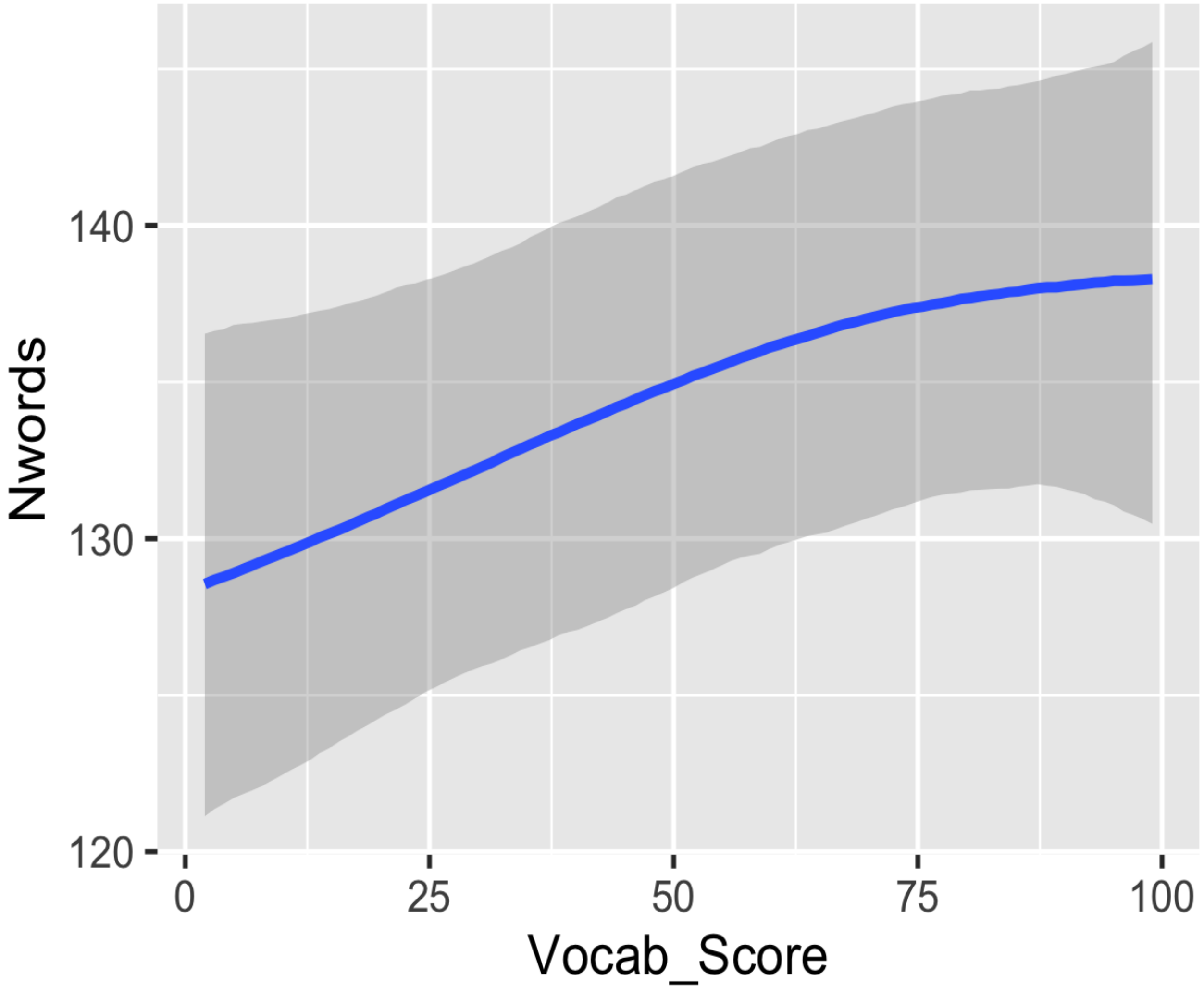
- Numeric variables may be in non-linear relationships with the response variable.
- To check that, one can use smooth terms, such as one-dimensional smooths `s()`, or two-dimensional terms (for two variables) `t2()`.
- We fit a model with a smooth term and inspect the plot:

```
mm_gam <- brm(Nwords ~ s(Vocab_Score) + University +
(1|Student_ID) + (1|Topic_ID), data = df, cores = 4,
control = list(adapt_delta = 0.99))
```

Conditional smooths

`conditional_smooths(mm_gam)`

No striking non-linearity.



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Illustrations

- Levshina, N., & Lorenz, D. (2022). Communicative efficiency and the Principle of No Synonymy: predictability effects and the variation of want to and wanna. *Language and Cognition*, *14*(2), 249–274. doi:10.1017/langcog.2022.7
- Levshina, N. (2022). Comparing Bayesian and Frequentist Models of Language Variation: The Case of Help + (to-)Infinitive. In O. Schützler & J. Schlüter (Eds.), *Data and Methods in Corpus Linguistics: Comparative Approaches* (pp. 224–258). Cambridge: Cambridge University Press.
- Thothathiri, M. & Levshina, N. (2023). Updating constructions: additive effects of prior and current experience during sentence production. *Cognitive Linguistics*, 34(3-4), 479-502. https://doi.org/10.1515/cog-2022-0020

Useful expressions

- *We fitted a Bayesian mixed-effects generalised linear model (GLMM) with the logit link function.*
- *The technical details about the Bayesian models and their goodness-of-fit measures are provided in the Supplementary Materials.*
- *The table displays the coefficients of the fixed effects and their 95% credible intervals. The right-hand column shows the posterior probability of the coefficients to be positive, computed as the proportion of positive values in the 6,000 posterior samples.*

Levshina & Lorenz 2022

Effects are strongly supported by the data

- *The coefficients and probabilities tell us that wanna occurs more often in conversations than in the other texts. The effect is very strong and highly credible, as we can see from the 95% credible intervals, which do not include zero in either of the models. Also, the posterior probabilities that conversations boost the chances of wanna are 100%.*
- *The presence of the negative particle decreases the chances of wanna with sufficient credibility…*
- *There is a clear effect of the speaker's sex in both models. Male speakers use wanna more often than female speakers. Also, the use of wanna is less likely in the older groups than in the younger groups.*

Levshina & Lorenz 2022

Lack of expected effects

- *There was no noteworthy effect of ObjectLonger (relative length of Object and Recipient).*
- *Critically however, we did not find an interaction between prior and current bias, which would reflect error-based learning. The model with the interaction term did not perform better than the more parsimonious model with respect to WAIC. When included, the interaction term for PO-only (compared to DO-only), which we were primarily interested in, was close to 0, with a mean posterior log-odds ratio of −0.30, 95 % credible interval between −1.11 and 0.50, and posterior probability of 22.5 %.*

Thothathiri & Levshina 2023

When 95% CI include zero but marginally

Note that the 95 % credible intervals include zeros, but only marginally. We have in both cases a higher than 90% posterior probability that the similarity measure has a negative effect on the chances of PO. Overall, these results are not 100 % conclusive but the evidence weighs in favour of our hypothesis (cf. Vasishth et al. 2013). This evidence can be used later for identification of potentially relevant variables for future studies of the dative alternation. Detection of such suggestive effects is one of the epistemological advantages of using Bayesian models (cf. Levshina 2022).

Thothathiri & Levshina 2023

Why Bayesian?

- *Bayesian modelling represents an attractive alternative to frequentist regression, which is also known as the maximum likelihood method. First, it allows the researcher to test the alternative hypothesis directly, rather than to test if the null hypothesis can be rejected. Second, it does not involve binary decisions based on p-values (Levshina & Lorenz 2022).*
- *Bayesian regression also allows one to use priors, and recycle results from previous research…*
- *In practice, however, the coefficient estimates of maximum likelihood models and those of Bayesian models are very similar. What differs is the process of inference. The posterior distributions based on MCMC enable us to compute 95% credible intervals for an estimate, where the parameter of interest (e.g., a regression coefficient) falls with the probability of 95%. We can also compute the probability that a certain estimate has a positive or negative effect on the use of wanna vs. want to… (Levshina & Lorenz 2022).*

Reporting MCMC

- *Models with four Markov chains were tried out in order to test convergence, which was acceptable. Because of some autocorrelation in the chains, for the final model we created one very long chain with 20,000 iterations (2,000 in the warm-up phase) and the thinning parameter of 3, which means that only every third posterior sample was considered. This produced the total of 6,000 posterior samples. Also, due to some divergent transitions, the adaptive parameter delta was increased to 0.99. This slowed down the Markov chain, but ensured that the posterior samples were valid. All R-hat values were 1.00, and the effective sample sizes were sufficiently large to produce reliable posterior distributions, which are summarised in Section 4.*

Levshina & Lorenz 2022

Discussing the priors

Bayesian inference involves the use of priors, which inform the model of our expectations before we look at the data. In our case, we chose to use regularizing, or weakly informative Cauchy priors centred around 0 with the scale 2.5, which is common practice in logistic regression, where regression coefficients greater than 7 or smaller than -7 are uncommon (at least, in datasets of the size like ours). These priors were applied to fixed-effects coefficients and standard deviations of the random effects. This helps to improve inference by pushing the coefficients somewhat towards realistic values without imposing subjective expectations (Nicenboim & Vasishth, 2016).

Levshina & Lorenz 2022

Goodness of fit

The models fit the data well. The first model based on the small sample with complete observations has the following goodness-of-fit measures: the Bayesian pseudo-R^2, which ranges from 0 (useless model) to 1 (perfect fit), was 0.48, with the 95% credible interval from 0.45 to 0.51. The C-index, which is traditionally used for evaluation of logistic models and which ranges from 0.5 to 1, is 0.909, with the 95% credible interval from 0.900 and 0.918. The second model based on the large dataset with imputed values fits the data somewhat better: Its Bayesian pseudo-R^2 is 0.54 (95% credible interval 0.53 – 0.56), and the C-index is 0.938 (95% credible interval 0.934 – 0.943).

Levshina & Lorenz 2022

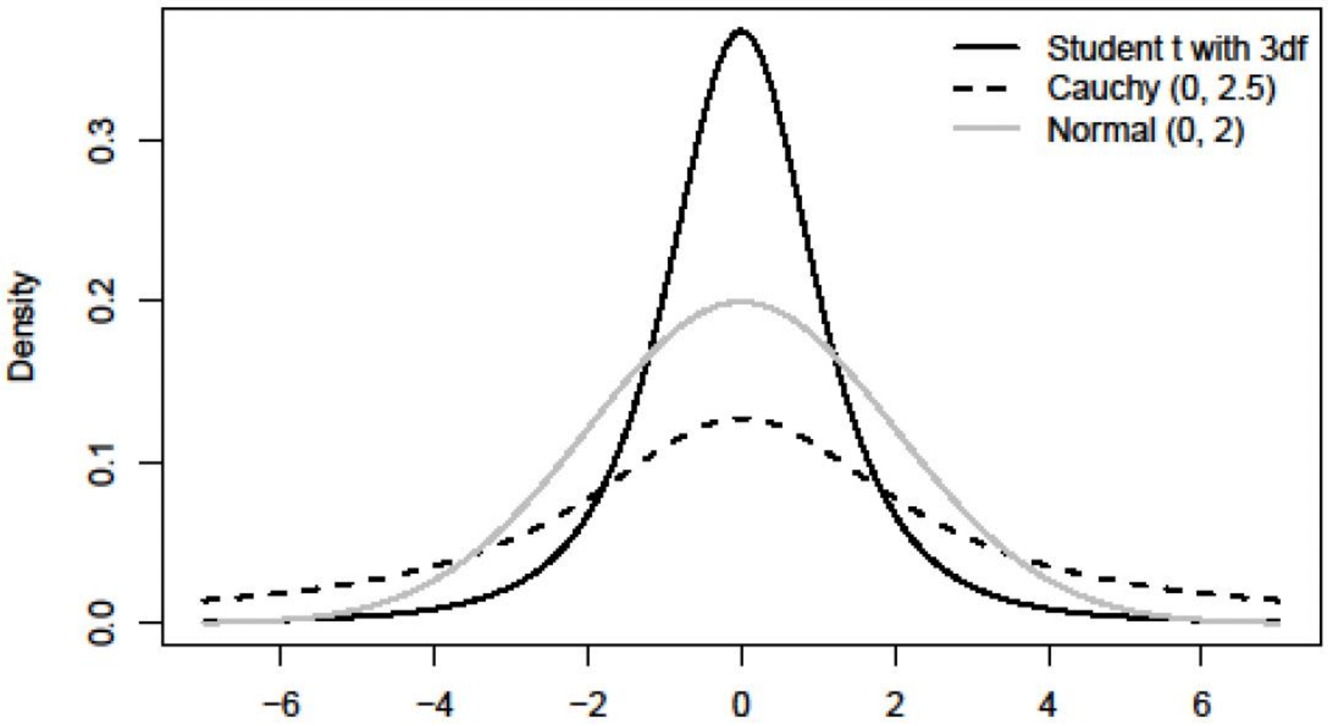
Prior sensitivity analysis

We can choose weakly informative priors based on well-known distributions, which are shown in Figure A1:

- *Student's t priors (e.g. with 3 degrees of freedom, centred around 0);*
- *Cauchy priors (e.g. centred around zero and with the scaling parameter of 2.5);*
- *Normal priors (e.g. with the mean of zero and the standard deviation of 2).*

These priors are weakly informative because they constrain the possible values only somewhat. Moreover, they do not specify the direction of an effect because they are centred around zero. One can see that the Cauchy priors have particularly fat tails, which means that they allow for more extreme values than the other priors.

Levshina 2022

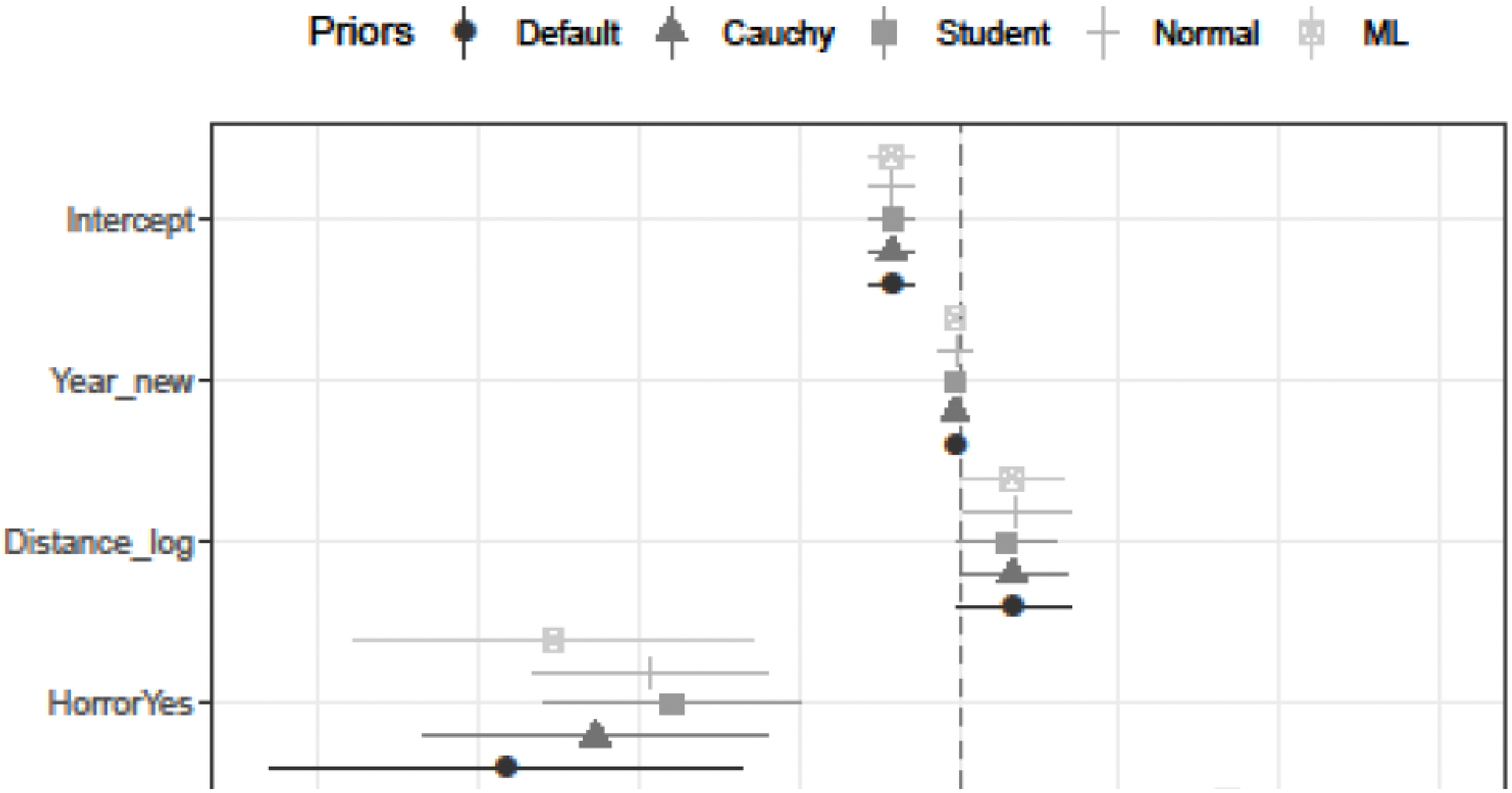


Levshina 2022

Let us fit models with these priors and see if the results differ. This is called sensitivity analysis. Each of the models contained all terms and interactions that were tested in the ML model. Four Markov chains were created, with 2000 steps in each. The burn-in (or warm-up) period was 1000 first observations. Therefore, we have in total (2000 – 1000) x 4 = 4000 posterior samples in each model.

Figure A2 displays the mean posteriors of the fixed effects in the four Bayesian models, and their 95% and 90% credible intervals. For the purposes of comparison, I also added the estimates of the ML model and their confidence intervals. One can see that the estimates and the intervals are very similar. Therefore, when we have a large dataset, the models with weak or flat priors are very similar (cf. Gelman et al. 2008), and closely approximate the results of an ML model. At the same time, there are some differences when the intervals are wide, as in the case of the coefficients related to horror aequi. The intervals are wide due to the fact that there are few observations with Horror=Yes and the to-infinitive as the response. In that case, the default flat priors model and the ML model have the most extreme values, being the least constrained.

Levshina 2022



Levshina 2022

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Why does MCMC get better with iterations?

- Efficient algorithms (e.g. Hamiltonian Monte Carlo as used by Stan/brms) have tuning parameters. That is, they learn to move more efficiently during a warm-up.
  - How big should the steps be?
  - They also learn the geometry of the posterior, so that they do not move equally in all directions.
- During the sampling phase (after the warm-up), the learning stops.
- This is why it's important to discard the warm-up iterations.

What is the role of adapt_delta?

- The no-U-turn sampler determines the step size by adapting it during the warm-up (burn-in) phase to a target acceptance rate (adapt_delta in Stan). The tuned step size is then used for all sampling iterations.
- High acceptance rate → the space is explored very slowly, in small steps. Your MCMC learns to walk carefully.
- Divergent transitions happen when NUTS cannot handle the parameter space correctly and fails. Such cases require more careful, smaller steps. Reasons: multicollinearity, sparse data, outliers…

What next?

The Bayesian being lifted by cranes during salvage operations. Photograph: Salvatore Cavalli/AP

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